Lecture Structure

Introduction & Hook

Ask students:

“Why does a VR environment look realistic when light and shadows are correct?”

Explain:

In VR realism depends heavily on **two things**:

1. Lighting (how objects interact with light)
2. Camera (how the user views the virtual world)

Without proper lighting and camera positioning, even high-quality 3D models look unrealistic.

PART 1: Lighting in VR

Importance of Lighting

Lighting determines:

- Object visibility
- Depth perception
- Surface detail
- Mood of environment

Example:

- Bright lighting → daytime scene
- Low lighting → horror or night scene

Lighting also helps users understand **shape and distance** of objects.

Lighting Models

Definition

A lighting model describes how light interacts with surfaces in a virtual scene.

It calculates color and brightness based on:

- Light source
- Surface properties
- Camera position

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1. Ambient Lighting

Simple background lighting applied uniformly.

Characteristics:

- No direction
- No shadows
- Simulates indirect light

Example:
Soft light in a room.

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2. Diffuse Lighting (Lambert Model)

Represents light scattered equally in all directions.

Depends on:

- Angle between light source and surface

Used for:

- Matte surfaces like walls or terrain.

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3. Specular Lighting

Represents shiny reflections.

Example:

- Metal
- Glass
- Water

Produces highlights on surfaces.

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4. Phong Lighting Model

Combines three components:

- Ambient
- Diffuse
- Specular

This is widely used in many graphics engines.

Lighting Equation Concept

Total light =
Ambient + Diffuse + Specular

This determines final pixel color.

PART 2: Cameras in VR

What is a Camera in VR?

Definition:

A camera represents the viewpoint through which the user sees the virtual environment.

In VR:

- Camera corresponds to the **user's head position**
- Movement of head updates camera position instantly.

This is called **head-tracked rendering**.

Types of Cameras

1. First-Person Camera

User sees the world from their own perspective.

Used in:

- VR games
- VR simulations
- Training environments

Provides maximum immersion.

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2. Third-Person Camera

Camera is placed behind the character.

Used in:

- Some VR games
- Character-based experiences

Less immersive than first person.

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3. Free Camera

Camera moves independently in the scene.

Used for:

- Development
- Debugging
- Scene exploration

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4. Stereoscopic Camera

Used in VR rendering.

Two cameras simulate human eyes:

- Left eye camera
- Right eye camera

This creates **depth perception**.

PART 3: Camera Movement in VR

Camera Movement Types

Head Tracking

Camera moves according to headset sensors.

Translational Movement

User walks or moves in virtual space.

Rotational Movement

User turns head left/right/up/down.

Important VR Rule

Camera movement must be **smooth and low-latency**.

Otherwise:

- Motion sickness
- Disorientation

Real Example

In VR training simulation:

- Lighting helps user see machinery clearly
- Camera moves as user turns head
- Stereoscopic cameras generate depth perception

Recap & Questions

Key Points

- Lighting determines realism in VR scenes
- Lighting models include ambient, diffuse, and specular
- Cameras define the user's viewpoint
- VR uses stereoscopic cameras for depth perception
- Smooth camera movement is essential for comfort

Quick Questions

1. What is the purpose of lighting models?
2. What are the three components of the Phong lighting model?
3. Why are two cameras used in VR?
4. What type of camera provides the highest immersion?

Additional Numericals for Lecture 21

Topic: Lighting Models in VR

Trignometery angles

COS	0	30	45	60	90
values	1	Root(3)/2	Root(2)/2	1/2	0

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1. Numerical: Ambient Lighting

Formula

Ambient lighting is the simplest model:

$I = I_a \times k_a$

Where:

- I = final intensity
- Ia = ambient light intensity
- ka = ambient reflection coefficient

Example Problem

Ambient light intensity = **0.8**
Ambient reflection coefficient of surface = **0.5**

Find the final intensity.

Solution

$I = 0.8 \times 0.5 = 0.4$

Final light intensity = 0.4

Explain to students:

The object reflects only part of the ambient light.

2. Numerical: Diffuse Lighting (Lambert Model)

Formula

$$I = Il \times kd \times \cos(\theta)$$

Where:

- Il= light source intensity
- kd = diffuse reflection coefficient
- θ = angle between light direction and surface normal

Example Problem

Light intensity = **1.0**
Diffuse reflection coefficient = **0.6**
Angle between light and surface = **60°**

Find the diffuse intensity.

Solution

$\cos(60°)=0.5$
$I=1.0\times0.6\times0.5$
$I=0.3$

Diffuse lighting intensity = 0.3

Explain:

As the angle increases, the surface receives less light.

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3. Numerical: Specular Lighting

Formula

$I = Il \times ks \times (\cos(\alpha))^n$

Where:

- ks = specular coefficient
- α\alpha = angle between reflection vector and viewer
- n = shininess factor

Example Problem

Light intensity = **1.0**
Specular coefficient = **0.7**
Angle = **45°**
Shininess factor = **2**

Solution

$\cos(45°)=1.414/2 = 0.707$
$(0.707)^2 \sim 0.5$
$I=1.0\times0.7\times0.5$
$I=0.35$

Specular intensity = 0.35

Explain:

Higher shininess makes highlights sharper.

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4. Numerical: Phong Lighting Model

Phong model combines all components:

I= ambient + diffuse + specular
$I= Ia \times ka + Ii \times kd \times \cos(theta) + Il\times ks\times(\cos(\alpha))^n$

Example Problem

Given:

Ambient intensity Ia=0.3
Ambient coefficient ka=0.5

Light intensity Il=1.0

Diffuse coefficient kd=0.4

Angle θ=60°

Specular coefficient ks=0.3

Angle α=30°

Shininess n=2

Find total intensity.

Step 1: Ambient

$0.3 \times 0.5 = 0.15$

Step 2: Diffuse

$\cos(60°) = 0.5$
$= 1.0 \times 0.4 \times 0.5 = 0.21$
$= 0.2$

Step 3: Specular

$\cos(30°) = 0.866$
(0.866)^2~0.75
$= 1.0 \times 0.3 \times 0.75 = 0.225$

Final Intensity

$I = 0.15 + 0.2 + 0.225$
$I = 0.575$
